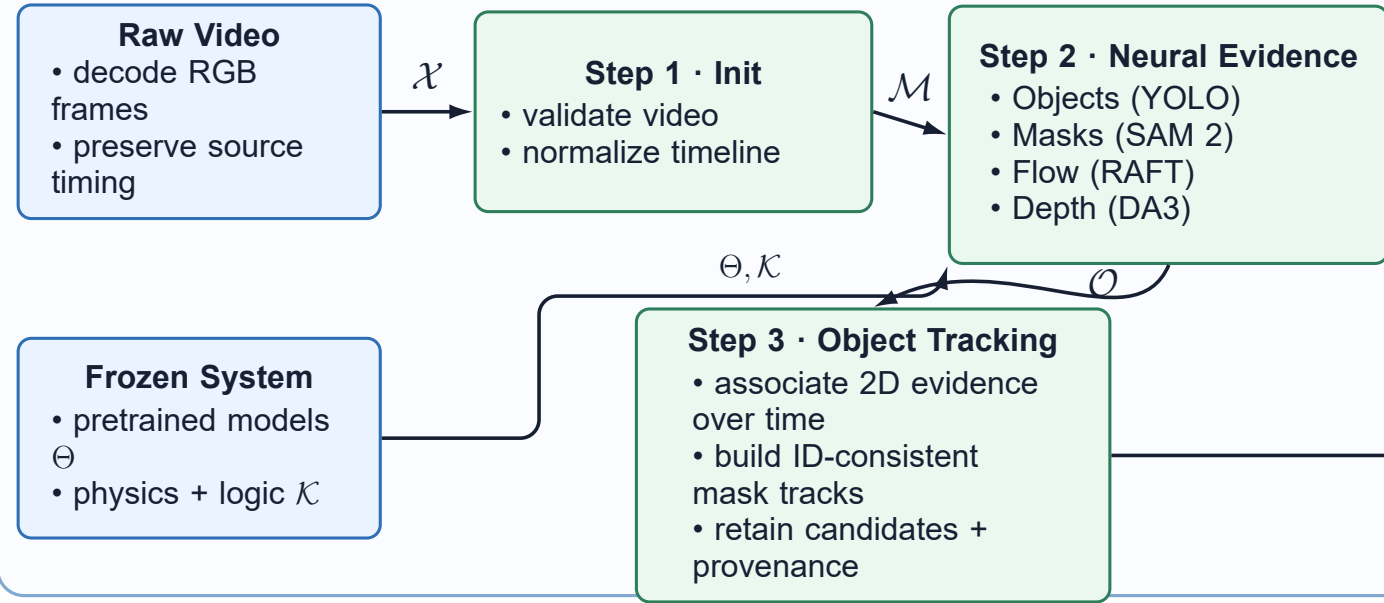


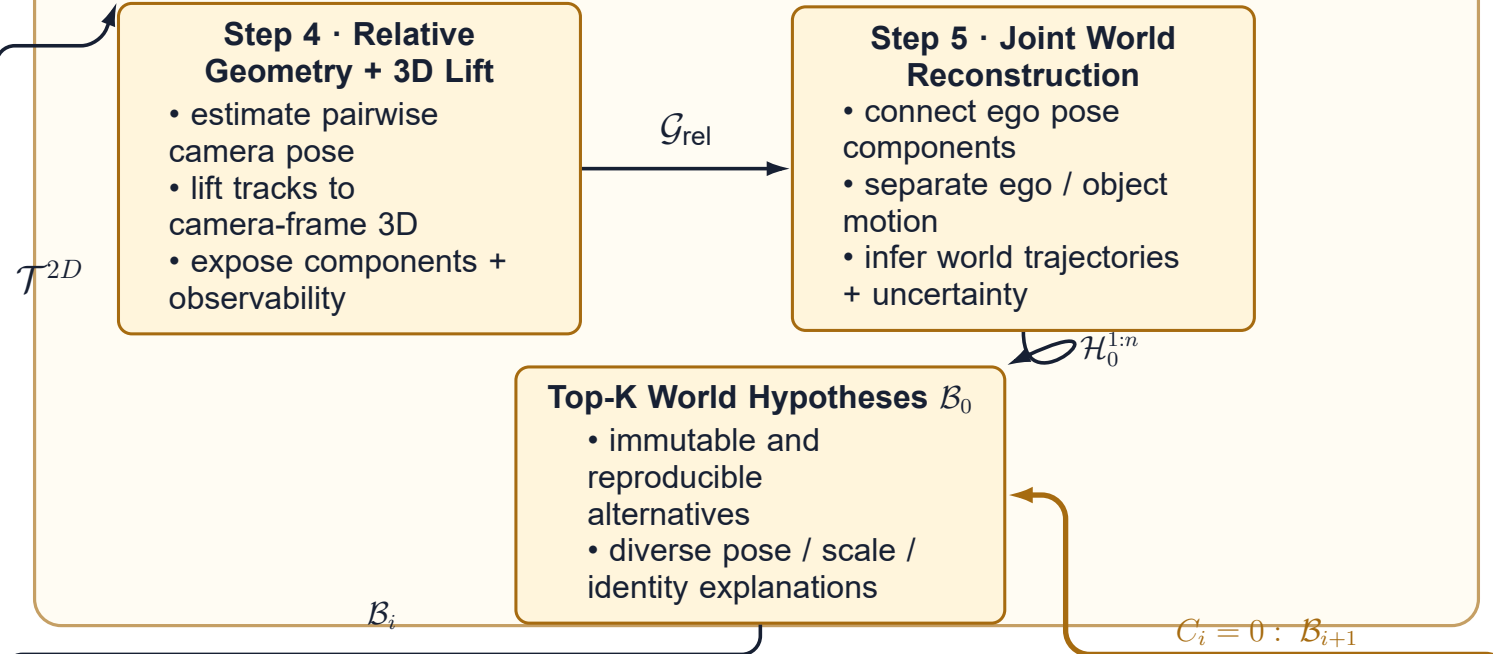
Training-free Evidence-grounded World-State Inference

Per-video inference only · no cross-video learning · annotations remain sealed until freeze

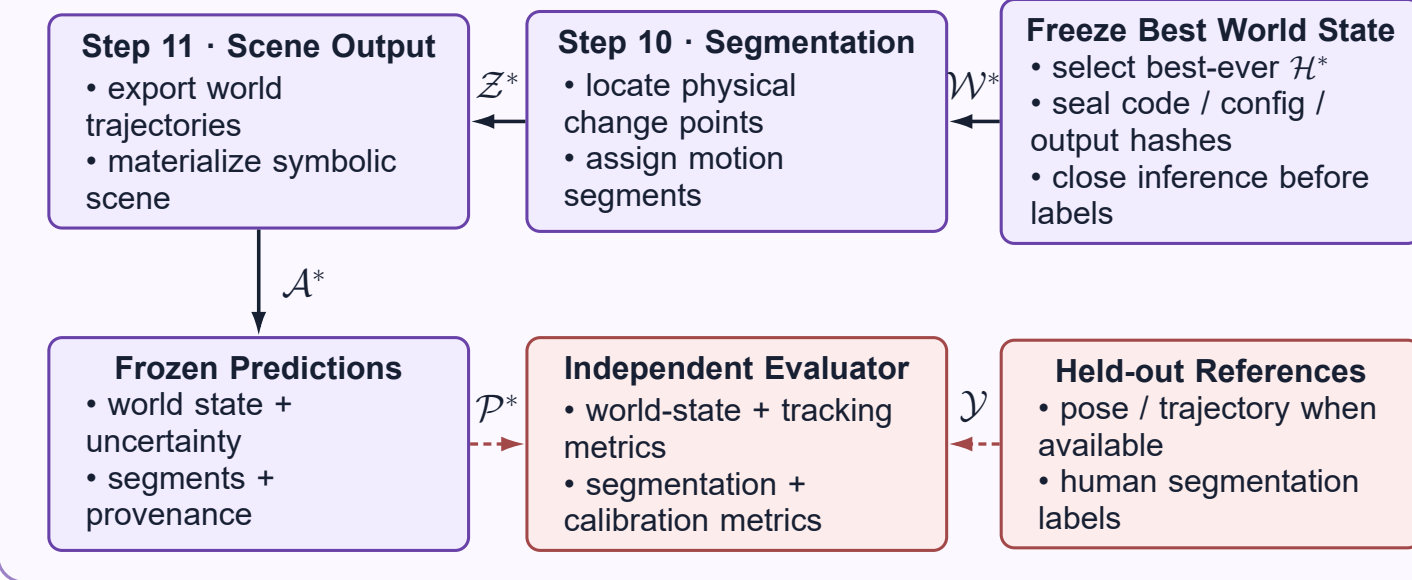
I · VIDEO → REPLAYABLE EVIDENCE



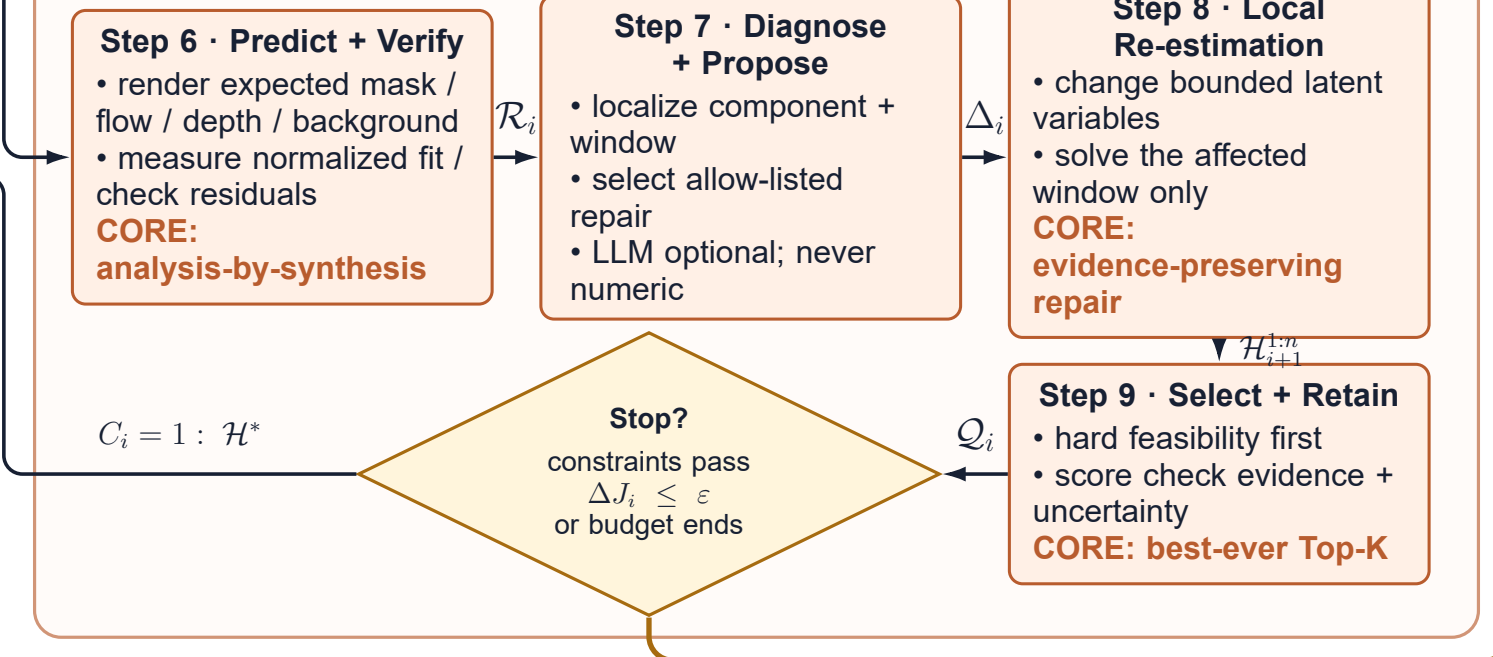
II · RELATIVE GEOMETRY → WORLD HYPOTHESES



IV · FREEZE → BLIND EVALUATION



III · ANALYSIS-BY-SYNTHESIS LOOP



$\mathcal{X}$ video · Θ frozen neural models · $\mathcal{K}$ frozen knowledge · $\mathcal{O}$ independent evidence · $\mathcal{T}^{2D}$ 2D tracks + archive · $\mathcal{G}_{rel}$ relative geometry + 3D observations

$\mathcal{B}_i$ world-state beam · Π fit/check roles · $\mathcal{R}_i$ residuals · Δ_i bounded repair · $\mathcal{H}^*$ best explanation · $\mathcal{W}^*$ frozen world · $\mathcal{Y}$ evaluation-only references